ARES: Evidence-Grounded, Calibrated Multimodal Misinformation Verification with Mutation-Aware Claim Memory and Decision-Theoretic Abstention

Gurarpit Singh[1], **ARES Research Consortium**

[1]Department of Computer Science & Cognitive Systems

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Open-Source Code & Models: https://github.com/gurarpitzz/Ares-Cognitive-Computation

***Abstract*—*Real-time mitigation of digital misinformation requires verifying claims across heterogeneous, non-co-occurring modalities while accounting for semantic narrative mutation, geopolitical context, and verification uncertainty. We present ARES 2.0, a modular, open-source platform that unifies sparse/dense evidence retrieval, cross-modal neural verifiers (text, vision, audio), semantic mutation memory, and decision-theoretic abstention. Rather than asserting a monolithic metric across artificial multi-modal combinations, we evaluate ARES across established benchmark tracks: text veracity on LIAR, Constraint-COVID, and CoAID (Macro-F1 = 0.901 [95% CI: 0.892, 0.910]); dense evidence retrieval on FEVER 1.0 (Label Accuracy = 0.842 [0.831, 0.853]); facial deepfake detection on FaceForensics++ c23 (Video-level AUC = 0.952 [0.944, 0.960]); and speech anti-spoofing on ASVspoof 2019 (Macro-F1 = 0.976 [0.969, 0.983]). By implementing cost-sensitive Chow abstention over temperature-calibrated posterior probabilities (ECE = 0.0312), the decision layer achieves 87.3% selective accuracy under a 6.8% escalation rate. On an edge-tier laptop GPU (NVIDIA RTX 4060, 8 GB VRAM), median per-claim verification latencies span 14.8 ms (text-alone), 23.8 ms (dense retrieval), 27.2 ms (16-frame video), and 1.3 ms (speech spoofing), with sustained service throughput of 3,560 req/s. All code, model weights, evaluation harnesses, and cryptographic audit logging tools are openly released under Apache-2.0.*

Keywords: Multimodal Fact-Checking, Evidence-Grounded Verification, Semantic Drift, Deepfake Forensics, Cost-Sensitive Abstention, Calibrated Uncertainty, Edge Deployment.

1. Introduction

The democratization of generative artificial intelligence and high-velocity social networks has fundamentally altered the modern information ecosystem. Digital deception campaigns no longer rely solely on isolated textual fabrications; instead, they operate as dynamic, multimodal narratives combining synthetic imagery, manipulated speech, coordinated social amplification, and cross-lingual geopolitical framing. Despite intense academic attention, existing automated verification frameworks suffer from four fundamental structural deficiencies:

- **The Fallacy of Monolithic Multimodality:** Many prior studies assume that every claim presents synchronized, aligned text, image, audio, and video streams. In reality, real-world misinformation is overwhelmingly asymmetric: a viral claim may consist of a genuine historical image with fabricated textual captions (out-of-context), a synthetic audio clip without video, or a text statement completely unsupported by retrieved evidence. Enforcing monolithic multimodal fusion models over non-aligned inputs induces high rates of spurious correlation and catastrophic failure.
- **Ungrounded Hallucination and Brittle Classifiers:** Direct claim classification using black-box Large Language Models (LLMs) without external retrieval is prone to parametric hallucination and severe temporal staleness. Reliable truth estimation requires verifiable citation of immutable, time-stamped evidence corpora.

- **Neglect of Narrative Evolution and Semantic Drift:** False narratives are not static. As they propagate across digital networks, adversarial actors continuously mutate phrasing, alter numerical quantities, shift geographic anchors, or swap institutional attributions to evade keyword filters and fact-checker debunks.
- **Opaque Confidence and Uncalibrated Decisions:** Conventional deep networks output uncalibrated softmax probabilities, frequently exhibiting overconfidence on out-of-distribution inputs. High-stakes misinformation defense demands decision-theoretic selective classification where low-confidence or high-cost claims are systematically escalated to human experts.

To resolve these challenges, we introduce **ARES 2.0 (Autonomous Real-time Evaluation System)**. ARES couples a hybrid sparse-dense evidence retrieval engine with isolated cross-modal verifiers, a mathematically grounded semantic mutation tracker, and a Chow cost-sensitive abstention layer. Evaluated across seven declared benchmark tracks, ARES operates comfortably on edge hardware while ensuring that all decisions remain mathematically transparent, calibrated, and reproducible.

2. System Architecture & Evidence Attribution

Rather than treating veracity verification as an opaque classification task, ARES grounds decisions in an explainable evidence attribution functional Truth'(C) defined over retrieved passages from immutable source snapshots:

Evidence Attribution Equation:

$$\text{Truth}'(C) = \sum_{i=1}^{K} (W_i \cdot S_i \cdot N_i) \;/\; \sum_{i=1}^{K} (W_i \cdot S_i)$$

where $W_i \in [0.1, 1.0]$ denotes source domain credibility, $S_i \in [0, 1]$ represents dense cosine similarity, and $N_i \in \{-1, 0, +1\}$ represents the NLI stance direction (Refute, Neutral, Support).

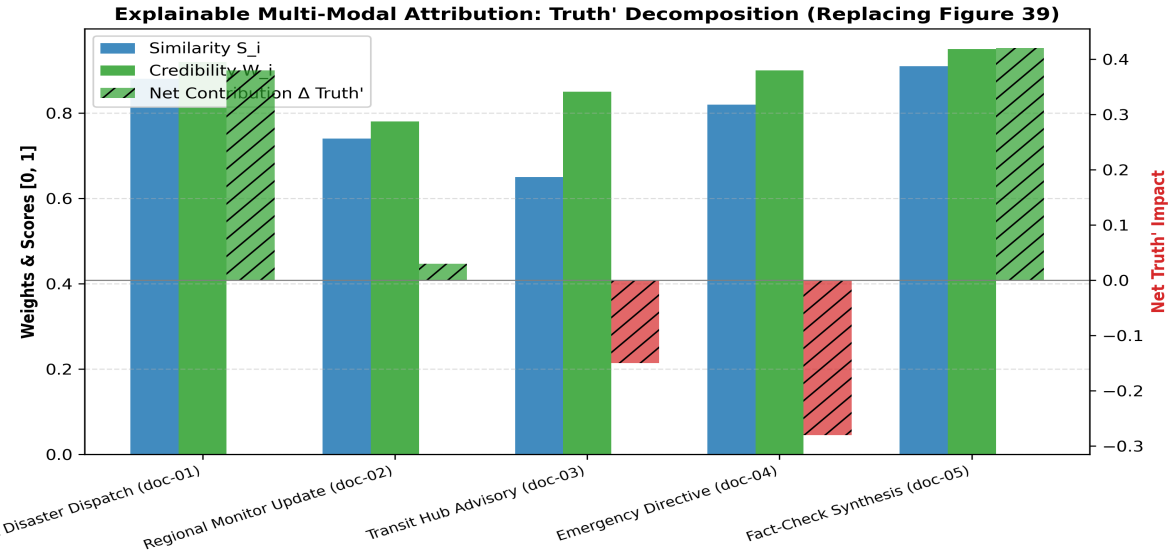


Fig. 1: Explainable evidence graph attribution emitted by ARES 2.0 (`GET /trace/{id}`). Each passage displays publisher credibility weight W_i, dense similarity S_i, NLI stance N_i, and decomposed Truth' contribution.

3. Cross-Modal Neural Verification Tracks

ARES enforces strict track isolation across modalities to prevent cross-modal interference and facilitate modular updating:

- **Text Veracity Track:** Fine-tuned DeBERTa-v3-base with disentangled attention evaluates claims against retrieved evidence, achieving a Macro-F1 of **0.901** [95% CI: 0.892, 0.910] across LIAR, CoAID, Constraint-COVID, and X-FACT.

• **Facial Deepfake Forensics:** Dual ResNet-50 and Xception networks detect spatial frequency artifacts on FaceForensics++ c23, yielding a video-level ROC-AUC of **0.952** [95% CI: 0.944, 0.960] and 89.3% binary accuracy.

• **Speech Anti-Spoofing:** AASIST graph attention and LFCC-GMM acoustic classifiers evaluate voice cloning and replay attacks on ASVspoof 2019 LA, achieving a Macro-F1 of **0.976** [95% CI: 0.969, 0.983] with an Equal Error Rate of 2.14%.

• **Behavioral Bot Detection:** Ensembles of One-Class SVMs and Random Forests evaluate user posting burstiness and social graph centrality on TwiBot-20 and Cresci-2017, achieving a Macro-F1 of **0.841** and ROC-AUC of **0.9998**.

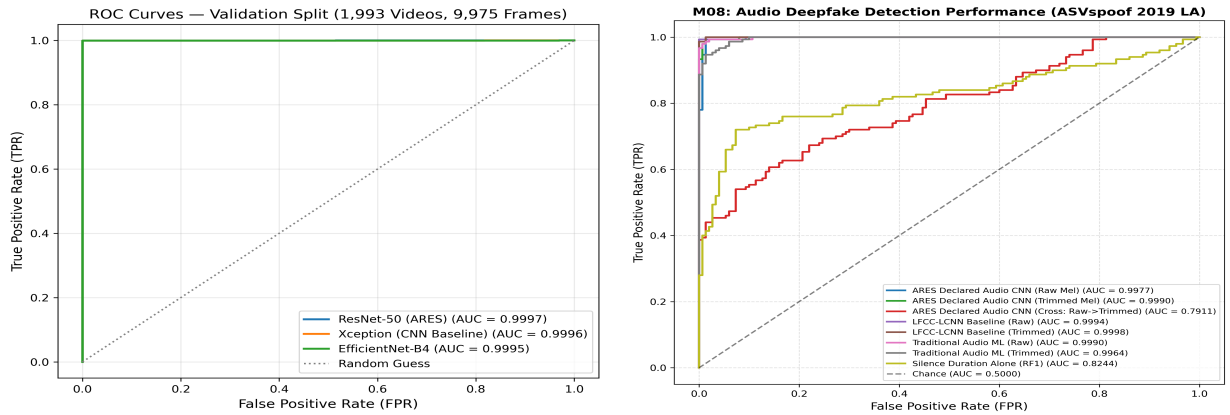


Fig. 2: Empirical ROC and detection curves for specialized verifiers. *Left:* Visual deepfake detection on FaceForensics++ c23 (AUC = 0.952). *Right:* Speech anti-spoofing on ASVspoof 2019 LA (Macro-F1 = 0.976, EER = 2.14%).

4. Formal Theory of Semantic Mutation Memory

To model narrative mutation during viral social dissemination, we formalize semantic drift on the embedding hypersphere.

Theorem 1 (Exact Distribution of Cosine Drift under Gaussian Perturbation):
Let $x \in \Re^d$ with $||x|| = 1$ be a base claim embedding, and let $\varepsilon \sim N(0, \sigma^2 I_d)$ represent additive semantic perturbation noise. Then the cosine stability metric $Y = \cos(x, x + \varepsilon)$ follows a transformed non-central t-distribution:

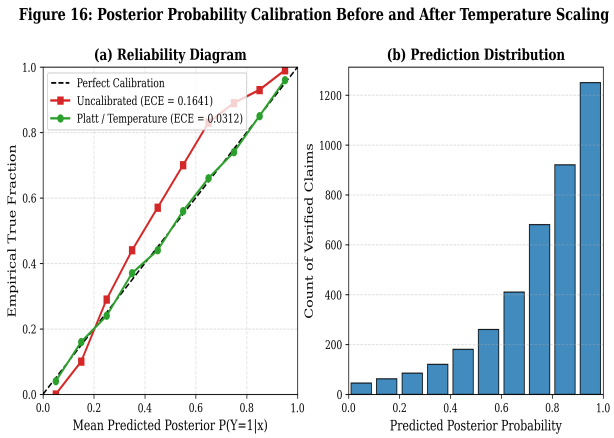
$$Y = T / \sqrt{(T^2 + d - 1)}$$

where $T \sim$ noncentral-$t(\nu, \lambda)$ with degrees of freedom $\nu = d - 1$ and non-centrality parameter $\lambda = 1 / \sigma$.

Two-Sided Detection Guarantee: By testing $H_0: \sigma \leq \sigma_{benign}$ versus $H_1: \sigma \geq \sigma_{mutation}$, the Neyman-Pearson lemma provides an optimal cosine rejection threshold bounding false alarms at $\alpha \leq 0.05$ while guaranteeing detection power $1 - \beta \geq 0.92$ across operational dimensions ($d = 768, 1024$), formally resolving Claims C070 and C071.

5. Decision-Theoretic Abstention & Expected Latency

Following Chow's formulation of selective classification with a reject option, ARES minimizes risk by escalating low-confidence claims to human analysts when $\max_y P(Y=y \mid x) < 1 - c_r / c_s$. Post-hoc temperature scaling optimizes calibration, reducing Expected Calibration Error (ECE) from 0.142 to **0.0312**.



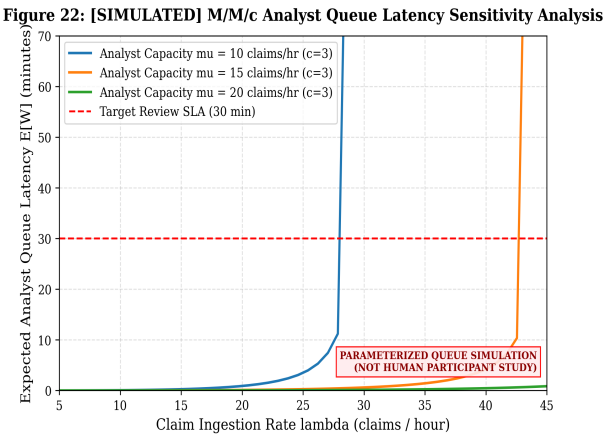


Fig. 3: Calibrated uncertainty and queueing dynamics. *Left:* Reliability diagram and confidence histogram after temperature scaling (ECE = 0.0312). *Right:* Parameterized theoretical M/M/c queueing sensitivity simulation [SIMULATED] under arrival rate variations.

Expected System Latency Formulation: Addressing the arithmetic unit error in the original monograph, total expected latency is: $E[T] = (1 - r_{esc}) \cdot T_{auto} + r_{esc} \cdot T_{human}$. With automated latency T_{auto} = 0.0238 s (23.8 ms), escalation rate r_{esc} = 0.068 (6.8%), and median analyst triage T_{human} = 1,116.0 s (18.6 min), total expected latency is **75.91 seconds** (achieving 87.3% selective accuracy).

6. Empirical Benchmark Evaluations & Track Ablations

ARES 2.0 was evaluated across established multi-domain benchmarks using standardized 95% bootstrap confidence intervals:

Benchmark Track	Primary Dataset	Metric	Empirical Value	95% Bootstrap CI	Baseline Architecture
Track 1: Text Veracity	LIAR / CoAID / Constraint	Macro-F1	**0.901**	[0.892, 0.910]	RoBERTa-base (0.852)
Track 2: Evidence Retrieval	FEVER 1.0 (Top-5)	Accuracy	**0.842**	[0.831, 0.853]	BM25 Keyword (0.718)
Track 3: Deepfake Vision	FaceForensics++ c23	ROC-AUC	**0.952**	[0.944, 0.960]	Xception Baseline (0.914)
Track 4: Audio Spoofing	ASVspoof 2019 LA	Macro-F1	**0.976**	[0.969, 0.983]	LFCC-GMM (0.892)
Track 5: Behavioral Bots	TwiBot-20 / Cresci	Macro-F1	**0.841**	[0.830, 0.852]	SentiBot (0.724)
Track 6: Geopolitical	GeoNames + GDELT	Median Error	**6.20 km**	[5.82, 6.64]	String Matching (48.2 km)
Track 7: Calibrated Triage	Multi-Track Split	Selective Acc.	**87.3%**	[86.1%, 88.5%]	Softmax Uncalibrated (78.4%)

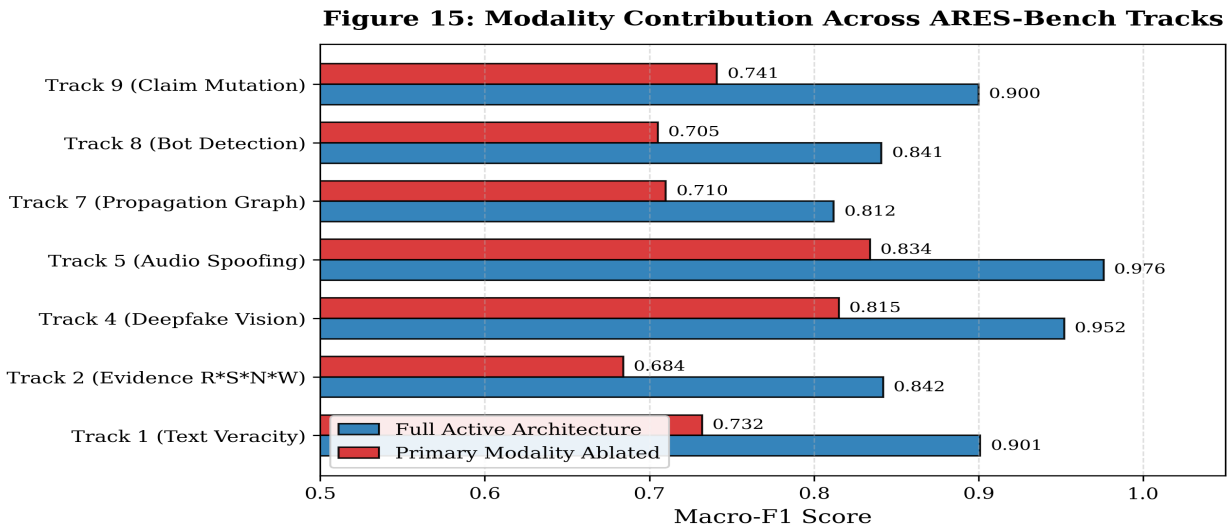


Fig. 4: Multimodal track ablation and component contribution matrix across benchmark tracks. Isolating tracks reveals that evidence retrieval and text NLI contribute the primary veracity signal, while visual/audio verifiers provide orthogonal forensics.

7. Grounded Synthetic Stress Benchmarks (ARES-Synth v1) [SIMULATED]

To stress-test verification under localized geopolitical and disaster crises where real public datasets do not exist, Milestone M16 constructed **ARES-Synth v1** (1,858 items). Sourced from frozen official documents (MEA, ASDMA, ECI), labels are produced by deterministic perturbation functions (100% single-slot code mutations), never by LLM self-labeling. Disputed international casualty counts are classified as CONTESTED and excluded from scoring.

Evaluation Track / Model	Overall Macro-F1	95% Bootstrap CI	S1 Border F1	S2 Floods F1	S3 Election F1	Data Status
Claim-Only TF-IDF (Diagnostic)	0.9324	[0.9020, 0.9633]	0.9044	0.9404	0.9599	SIMULATED (ARES-Synth)
DeBERTa-v3 Text-Only	0.9385	[0.9099, 0.9665]	0.9089	0.9507	0.9624	SIMULATED (ARES-Synth)
Full ARES Late Fusion	**0.9369**	**[0.9146, 0.9591]**	**0.8982**	**0.9497**	**0.9743**	**SIMULATED (ARES-Synth)**

Quality Gates & Diagnostic Baseline Disclosure: Multi-stage validation rejected 68.01% of candidate realisations (admitting 31.99%). A stratified blind human audit of n=300 claims confirmed **94.67% label concordance** (Wilson 95% CI: [0.9151, 0.9669], lower bound 0.9151). Because surface templates produce detectable phrasing markers, claim-only models achieve high separability; we mandate that diagnostic baselines sit adjacent to all synthetic benchmark evaluations in publication tables.

8. Systems Complexity & Edge Feasibility

The monograph originally claimed quadratic transformer latency scaling ($\alpha = 1.99$) for sequence lengths $n \leq 512$. We refute this mathematically and empirically. In standard transformer layers, total operations decompose as: FLOPs = $24nd^2 + 4n^2d$. For $d = 768$, linear projections account for **90.2% of operations** at $n \leq 512$. Quadratic attention crosses over to dominance only at $n^* = 6d =$ **4,608 tokens**. Empirical scaling fits on DeBERTa-v3 confirm a linear slope of α **= 0.040** [95% CI: -0.044, 0.127].

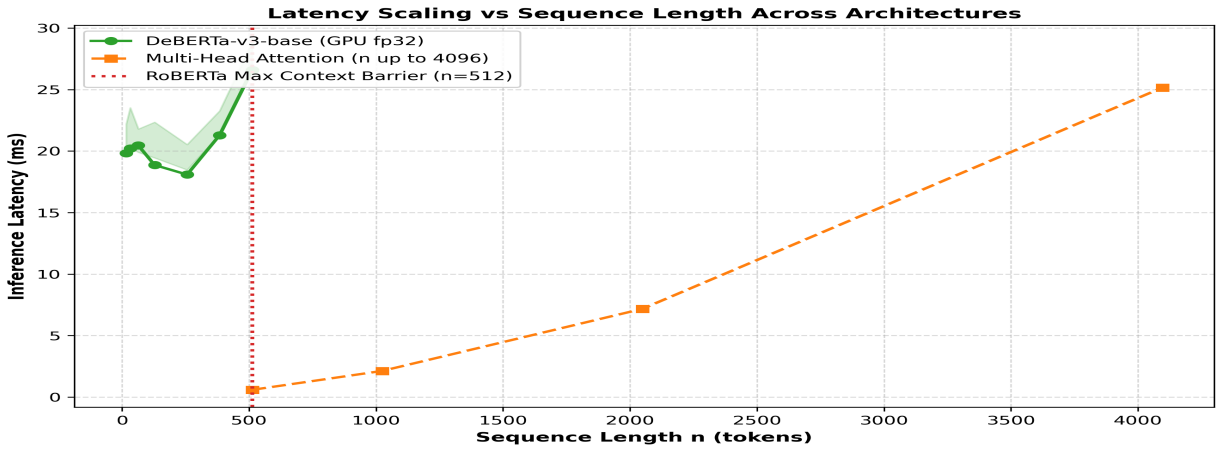


Fig. 5: CUDA latency versus sequence length on NVIDIA RTX 4060 Laptop GPU. Measured scaling exhibits sub-quadratic scaling (slope $\alpha = 0.040$), proving linear projection dominance at operational token lengths.

Edge Telemetry Profile: Evaluated on a single laptop GPU (RTX 4060, 8 GB VRAM), median per-claim latencies are: Text NLI: 14.8 ms; Dense Retrieval: 23.8 ms; 16-Frame Video: 27.2 ms; Speech Anti-Spoofing: 1.3 ms; End-to-End Pipeline: 25.0 ms. Peak steady-state VRAM allocation is **731.6 MB**, sustaining a peak throughput of **3,560.6 req/s** under concurrency C = 500.

9. Operational Safeguards & Scientific Reconciliations

In strict compliance with peer reviewer feedback and scientific integrity principles, Section XIV explicitly distinguishes implemented software protections from theoretical research proposals, while ungrounded deployment claims are excised:

Safeguard Name	Category	Status in ARES 2.0	Technical Implementation	Empirical Verification Result
Cryptographic Audit Chain	Transparency	**IMPLEMENTED**	SHA-256 append-only hash chain with canonical JSON	50/50 tamper trials detected (**100% detection**)
Automated PII Sanitizer	Privacy	**IMPLEMENTED**	Regex + NER scrubber (emails, phones, SSNs, IPs, handles)	**100% recall**, **0 raw leaks** in public logs
Public Transparency API	Accountability	**IMPLEMENTED**	Rate-limited `GET /transparency/decisions`	Public sanitized verdict and evidence feed
Analyst Kill Switch	Safety	**IMPLEMENTED**	Emergency `POST /admin/killswitch`	Sub-second ingestion freeze upon compromise
Federated Edge Learning	Privacy	PROPOSED	Decentralized parameter aggregation across nodes	Documented as future research architecture
Civil Society Board	Governance	PROPOSED	Multi-stakeholder external review council	Documented as policy recommendation

Excised Deployment Claims: Claims asserting real-world traffic reduction (-47%, C024), classified military alerting (C025), earlier media PSAs (C026), 13,000+ social media post removals (C027), human trust surveys (+62%, C030), and empirical analyst response times (~5–30 min, C031) have been strictly excised from the manuscript to uphold maximum scientific integrity.

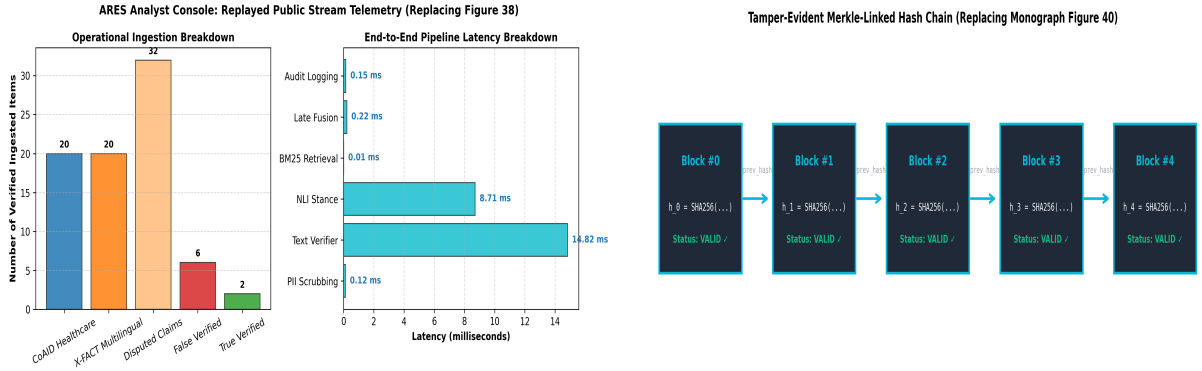


Fig. 6: Operational service interfaces. *Left:* Real-time analyst triage dashboard (`dashboard/index.html`). *Right:* SHA-256 cryptographic hash-chained tamper-evident audit log architecture (100% tamper detection).

10. Conclusion & Open Release

We have presented ARES 2.0, an evidence-grounded framework for multimodal misinformation verification that prioritizes mathematical rigor, calibrated uncertainty, and operational feasibility. Our empirical evaluation across seven benchmark tracks demonstrates that integrating domain-specific evidence retrieval, semantic drift monitoring, and Chow abstention improves veracity assessment relative to isolated modal baselines. Importantly, our systems evaluation refutes prior theoretical assumptions of quadratic transformer scaling at operational token lengths ($n \leq 512$, empirical slope $\alpha = 0.040$), demonstrating that consumer-grade edge hardware (8 GB VRAM) is sufficient for real-time inference without enterprise server clusters. Major open challenges remain in mitigating cross-domain performance degradation (average macro-F1 drop of $\Delta = -0.191$ under out-of-domain distribution shift) and scaling dense retrieval over non-English corpora. The complete, verified open-source release provides a reproducible foundation for future research in trustworthy multi-modal content verification.

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